

Non-Line-of-Sight Imaging based on Dual Photography using Leaked EM Waves

Masaya Oishi
24wm3270@student.gs.chiba-u.jp
Chiba University
Japan

Taiki Kitazawa
Yuichi Hayashi
kitazawa.taiki.kq8@is.naist.jp
yu-ichi@is.naist.jp
Nara Institute of Science and
Technology
Japan

Hiroyuki Kubo
hkubo@chiba-u.jp
Chiba University
Japan

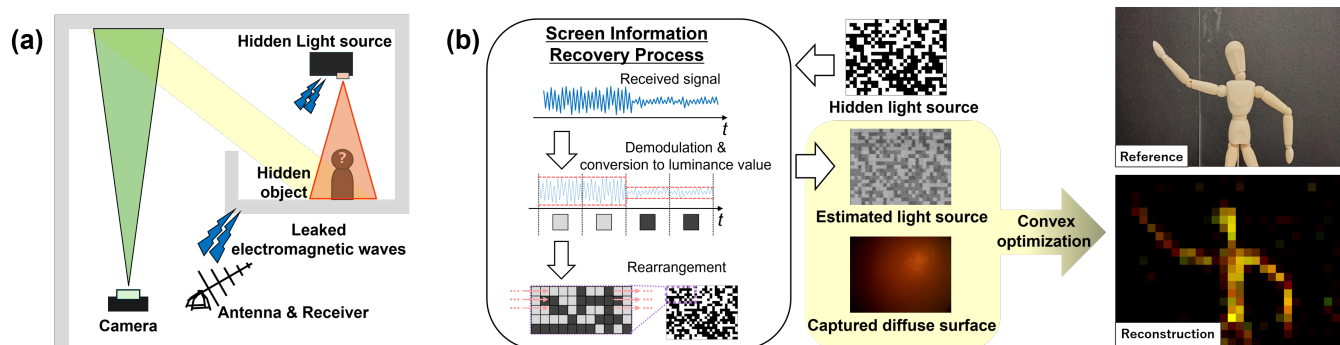


Figure 1: An experimental setup to reconstruct NLoS hidden object by estimating an unknown projected light pattern and the diffuse surface of the wall illuminated by the reflected light of the hidden object (a). In our method, we first estimate the projected light pattern by EM waves leaked from the projected device. Then, we capture the diffuse surface of the wall. Based on Helmholtz reciprocity, we achieve NLoS images through convex optimization (b).

ABSTRACT

In this study, we propose a novel method for Non-Line-of-Sight (NLoS) imaging utilizing leaked electromagnetic (EM) waves. Our approach leverages the ability to estimate images from projected by a hidden light source (projector) from leaked EM waves. By estimating the projected pattern from a hidden light source and capturing the diffuse surface of a wall illuminated by the reflected light of the hidden object, we demonstrate the reconstruction images of objects occluded from direct view. We show that leaked EM waves can be exploited to enhance NLoS imaging techniques, opening up new possibilities for reconstructing scenes beyond the direct field of view.

CCS CONCEPTS

• Computing methodologies → Computational photography.

KEYWORDS

Non-Line-of-Sight imaging, leaked electromagnetic waves, linear inverse problem

ACM Reference Format:

Masaya Oishi, Taiki Kitazawa, Yuichi Hayashi, and Hiroyuki Kubo. 2024. Non-Line-of-Sight Imaging based on Dual Photography using Leaked EM Waves. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641234.3671030>

1 INTRODUCTION

In the field of computational imaging, researchers have developed various techniques to visualize NLoS scenes. The key advancements in this area include the use of Single-Photon Avalanche Diode (SPAD) sensors [O'Toole et al. 2018]. In contrast, we propose a novel NLoS imaging method that utilizes leaked EM waves captured by an antenna, which are emitted from a hidden light source such as a projector. As shown in Figure 1, we capture the diffuse surface of the wall illuminated by the reflected light of the hidden object. Simultaneously, we also capture the leaked EM waves to estimate the projected pattern from a hidden projector. Through the analysis of the linear inverse problem related to light propagation, we reconstruct images from the perspective of the light source. To the best of our knowledge, this is the first study to demonstrate NLoS imaging based on leaked EM waves.

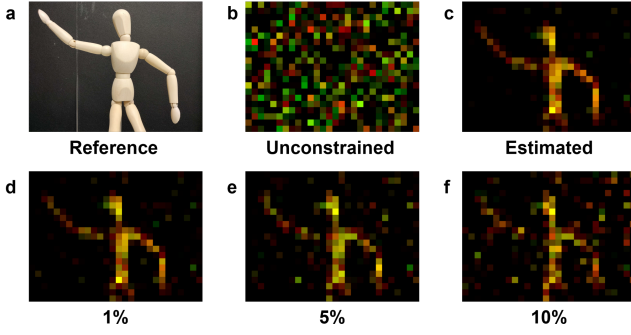


Figure 2: (a)The Reference Image of hidden object.(b)Due to inaccuracies in the estimated information of hidden light sources, correct reconstruction results cannot be obtained with unconstrained pseudo inverse matrices.(c)The correct reconstruction results of the hidden object obtained by performing constrained convex optimization using the estimated hidden light source information.To examine the tolerance level of estimation errors in the hidden light source information within this method, intentional errors were introduced into the light source information. The reconstructions were performed with 1% noise of the total in (d), 5% in (e), and 10% in (f) of the total.

2 OUR METHOD

In our research scenario, a hidden object and an uncontrollable hidden light source are located in a NLoS position, as observed by a camera and an antenna, as illustrated in Fig. 1(a). This setup is similar to the one presented in dual photography [Sen et al. 2005]; however, the key difference is that the light source is hidden and cannot be controlled by the observer. In this method, we estimate the projection pattern by analyzing unintentionally leaked EM (EM) waves emitted during the operation of the projector [Kitazawa et al. 2023]. By continuously measuring the projected pattern p' , estimated from the leaked EM waves, and the diffuse surface of the wall illuminated by the reflected light of the hidden object c' , the image of the NLoS object can be reconstructed. This is achieved by analyzing the light transport matrix T derived from the relationship between p' and c' , similar to the approach used in dual photography [Sen et al. 2005]. However, the images estimated from leaked EM waves contain estimation errors, which makes reconstruction challenging without any regularizer to solve the equation. To address this issue, we adopted convex optimization with regularization terms as:

$$\begin{aligned} \underset{T \geq 0}{\operatorname{argmin}} \quad & (\|Tp' - \bar{c}'\|_2^2 + \alpha E_c + \beta E_s) \\ \bar{c}' = \quad & \frac{1}{n \times m} \sum_{i=1}^{n \times m} c'_i \\ E_c = \quad & \left\| \frac{\partial}{\partial x} T + \frac{\partial}{\partial y} T \right\|_2^2, \quad E_s = \|T\|_1. \end{aligned} \quad (1)$$

Let T denote the light transport matrix, p' the estimated projected pattern, and $\bar{c}'$ the average pixel value of each captured image.

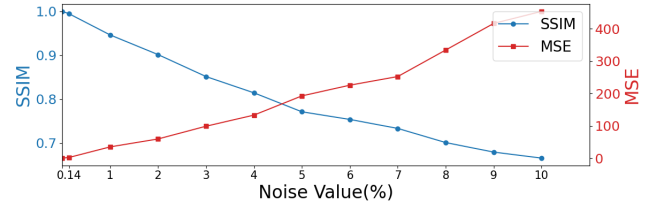


Figure 3: The results of the reconstruction evaluation, measured by SSIM and MSE, indicate that an estimation error of approximately 4% can be tolerated.

3 EXPERIMENT AND RESULT

The experimental setup consisted of an IDS UI-3250CP camera, a Sony MP-CL1A projector, and a National Instruments B210 for measuring leaked EM waves. To simplify the problem formulation, we assumed that the images projected from the hidden light source were binary, with black and white pixels, and had a resolution of 28×21 pixels. A total of 588 captures and estimations were performed. In this experiment, the receiving frequency (f_c) was set to 371 MHz, as the obtained signal varies depending on the central frequency used for measuring leaked EM waves. Under this assumption, the estimated light source information achieved sufficient accuracy of 99.86%. However, when estimating images with higher resolutions, the estimation accuracy may decrease. To investigate this, intentional bit flips were performed on 1-10% of the total elements, and the reconstruction results are presented in Fig. 2. To further investigate the loss of linearity caused by estimation errors resulting from leaked EM waves, we also present results obtained using an unconstrained pseudo-inverse matrix without regularization in Fig. 2. The robustness of the proposed method against noise was evaluated by calculating the structural similarity index measure (SSIM) and mean squared error (MSE), as shown in Fig. 3.

4 CONCLUSION

In this paper, we propose a novel approach to NLoS imaging that utilizes leaked EM waves. As a future work, we extend the applicability of this method to scenarios where displays, rather than projectors, serve as the light sources.

ACKNOWLEDGMENTS

This work was sponsored by the JSPS KAKENHI grant number JP24K00783, Grant-in-Aid for JSPS Fellows JP23KJ1593, and JST FOREST JPMJFR206L, JPMJFR206L.

REFERENCES

- Taiki Kitazawa, Hiroyuki Kubo, and Yuichi Hayashi. 2023. A Method for Extracting Plausible Images From EM Leakage Measured at Low Sampling Rates. *IEEE Letters on Electromagnetic Compatibility Practice and Applications* (2023).
- Matthew O'Toole, David B. Lindell, and Gordon Wetzstein. 2018. Confocal Non-Line-of-Sight Imaging Based on the Light-Cone Transform. *Nature* (2018).
- Pradeep Sen, Billy Chen, Gaurav Garg, Stephen R Marschner, Mark Horowitz, Marc Levoy, and Hendrik PA Lensch. 2005. Dual photography. In *ACM SIGGRAPH 2005 Papers*. 745–755.